12. Earley Parser for Context-Free Grammar

Aim: To implement an Earley parser for a context-free grammar using Python.

Procedure:

1. Install NLTK using python -m pip install nltk.
2. Define the grammar.
3. Create an Earley chart parser.
4. Parse the input sentence.
5. Display the parse trees.

Code:

```
import nltk

grammar = nltk.CFG.fromstring("""

S -> NP VP

NP -> Det N

VP -> V

Det -> 'the'

N -> 'boy'

V -> 'runs' """)

parser = nltk.EarleyChartParser(grammar)

sentence = "the boy runs".split()

for tree in parser.parse(sentence):

    print(tree)
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/12.py"
(S (NP (Det the) (N boy)) (VP (V runs)))
```

Result: The Earley parser successfully parsed the sentence according to the context-free grammar.